

Machine learning-aided optimization for transport layer parameters of low lead inorganic Zn-based mixed-halide perovskite solar cell

Navdeep Kaur^a, Rahul Pandey^{a,*}, M. Khalid Hossain^b, Jaya Madan^{a,*}

^a VLSI Centre of Excellence, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab 140401, India

^b Institute of Electronics, Atomic Energy Research Establishment, Bangladesh Atomic Energy Commission, Dhaka 1349, Bangladesh

ARTICLE INFO

Keywords:

Perovskite solar cell
SCAPS-1D
Machine learning
SHAP analysis

ABSTRACT

Perovskite-based solar cells have captivated researchers, due to their outstanding photovoltaic (PV) performance. All inorganic halide perovskite solar cells (AIHP-SC) exhibited excellent stability against environmental conditions with improved lifetime. In this investigation, low lead CsPb_{0.625}Zn_{0.375}I₂Cl based AIHP-SC is designed with the utilization of a SCAPS-1D simulator. Further, the impact of ETL/HTL parameters viz. mobility and doping on the PV performance is analyzed. Thereafter, machine learning (ML) models are trained, tested, and verified using artificial intelligence (AI) algorithms. Utilization of ML models shortens experiment time and eliminates the requirement for extensive resources in designing and predicting the PV performance of solar cells. In this research, the PV performance of AIHP-SC layered as TiO₂/CsPb_{0.625}Zn_{0.375}I₂Cl/Spiro-MeOTAD at different ETL/HTL mobility (ranging from 0.004 cm²/Vs to 4 cm²/Vs) and doping density (ranging from 10¹⁵ cm⁻³ to 10¹⁹ cm⁻³) has been evaluated. The influence of these variations leads to the generating 2500 PV performance datasets including current density (J_{SC}), fill factor (FF), open circuit voltage (V_{OC}), and power conversion efficiency (PCE). These generated data set attributes are fed to train the ML models viz. linear regression (LR), random forest (RF), support vector regression (SVR), eXtreme gradient boosting (xGB), and artificial neural network (ANN). For validation, the performance of ML models is verified against SCAPS-1D generated performance results by using mean square error (MSE) and R square (R²) as the performance metrics. Execution of ML algorithms revealed that performance of RF and xGB exhibits high correlation with SCAPS-1D generated data set as the prediction made by these two algorithms are best matched to actual outcome. Additionally, Shapley additive explanations (SHAP) analysis is performed to examine the influence of input variables (also known as independent variables) on target performance parameters (V_{OC}, J_{SC}, FF and PCE). A significant rise in PCE is observed from 9.5 % to 21.90 % while optimizing ETL/HTL mobility and doping density of AIHP-SC. The investigated results provide supervision to researchers in the design of highly stable AIHP-SC, further integration of ML in AIHP-SC is time time-efficient approach to design and performance prediction.

1. Introduction

In today's world, there is a growing emphasis on green energy and sustainability. One promising source of renewable energy is solar power, which is generated through the use of photovoltaic cells. By harnessing solar energy, we can reduce our reliance on fossil fuels and help mitigate the negative environmental impacts associated with traditional energy sources. Selection of appropriate material is the foremost step, in solar cell design. A few important material characteristics such as energy band gap, absorption coefficient, exciton binding energy, carrier lifetime and carrier mobility are instrumental in determining the suitability of a

material for solar cell design. Perovskite materials have emerged as a promising alternative to silicon, for designing photovoltaic cells [1,2]. High efficiency, versatility in band gaps, high absorption coefficient, longer diffusion length, and carrier lifetime, are the exceptional features of perovskite materials, which have made them one of the promising materials for solar cells and optoelectronic devices [3–5]. The perovskite material utilized in the absorber layer plays a crucial role in determining the electrical and optical properties of a solar cell.

Methylammonium lead iodide (CH₃NH₃PbI₃) is the most extensively studied hybrid perovskite material in optoelectronic applications [6–8]. In 2006, the first methylammonium lead (II) tribromide (CH₃NH₃PbBr₃)

* Corresponding authors.

E-mail addresses: rahul.pandey@chitkara.edu.in (R. Pandey), jaya.madan@chitkara.edu.in (J. Madan).

<https://doi.org/10.1016/j.solener.2023.112055>

Received 4 August 2023; Received in revised form 19 September 2023; Accepted 22 September 2023

Available online 26 September 2023

0038-092X/© 2023 International Solar Energy Society. Published by Elsevier Ltd. All rights reserved.

was introduced with 2.2 % PCE [9]. With an improved PCE of 3.8 %, methylammonium lead (II) triiodide ($\text{CH}_3\text{NH}_3\text{PbI}_3$) was designed in the year 2009 [10]. By the year 2022, EPFL institute claimed that the PCE of single junction hybrid perovskite solar cells (PSC), utilizing formamidinium lead triiodide (FAPbI_3), has elevated to 25.6 % [11]. A remarkable improvement in PCE with optimum optical properties in hybrid PSC attracted the attention of researchers, to further continue research in perovskite. MaSnI_3 and FASnI_3 are lead-free approaches in the design of hybrid PSC, in which the content of lead (Pb) is substituted with tin (Sn) [12,13]. FASnI_3 hybrid perovskite is highly efficient and offers more stability against temperature variations in contrast to MaSnI_3 [14]. Since organic cations present in hybrid perovskite material structures are susceptible to temperature variation, air and humidity that leads to degradation of atomic structure and consequently optical performance of material [15–18]. Researchers are currently exploring alternative materials that exhibit strong optical performance as well as longer shelf life. By replacing organic cations such as methylammonium and formamidinium with inorganic components such as cesium, all-inorganic perovskite solar cells (CsPbX_3) have emerged with superior performance in photovoltaic (PV) technology [19]. All inorganic CsPbX_3 yields significant improvement in stability as compared to hybrid-PSC, with outstanding light harvesting properties [19–24]. Further, it possesses excellent photoelectric properties such as long carrier lifetime, diffusion length and improved defect tolerance [25,26]. CsPbBr_3 , CsPbI_2Br , $\text{CsPbI}_2\text{Br}_2$, and CsPbI_3 are the most commonly adopted all-inorganic perovskite material compounds in design of solar cell [27,28]. The highest reported PCE of CsPbI_3 is 21.15 % with energy band gap of 1.7 eV [29]. However phase instability is still a challenge for CsPbI_3 in its commercial growth, different strategic approaches are being deployed to enhance stability [24]. CsPbCl_3 has a large energy bandgap, which causes the low absorption of light wavelengths, resulting in low PCE [30]. Moreover, high recombination and energy loss during charge transport, limit PV performance of CsPbBr_3 [31]. The presence of lead content in perovskite material spreads toxicity in the environment. Its determinant effect has an impact on the health of living beings, which enter the food chain through soil [32]. A total replacement of lead can degrade the PV performance of the solar cells. Therefore, approaches are being devised by researchers to reduce some portion of lead from the perovskite compound, while preserving its optical performance [33,34]. The modification in chemical structure of ABX_3 perovskite compound, helps to optimize the PV performance of PSC [35–37]. The bandgap of material can be adjusted by altering X-site ions of perovskite material [38]. In addition, variation in doping of B site metal ions (Zn^{2+} , Sr^{2+} , Mn^{2+} , Cu^{2+} , Sn^{2+} and Ge^{2+}), modify the PV performance of PSCs [39]. The trade-off between presence of lead in perovskite and its optical performance can be achieved by substitution of B site ions. This substitution of Zn with Pb in all inorganic material has shown noteworthy improvement in optical performance of perovskite material, includes improvement in thermal stability, reduced trap density and charge recombination [40–42].

In this research work, a low lead all inorganic $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ perovskite material is investigated, which is deployed as an absorber layer in AIHP-SC [43]. The optical performance of iodide-based all-inorganic perovskite, doped with chloride ions, demonstrates notable improvement in PV parameters [44–46]. In the aforementioned cesium-based perovskite material, 37.5 % of lead is reduced, and a significant amount of zinc is added, additionally, halide is doped with chloride [43]. The proposed design incorporating the abovementioned material composition variation in an all-inorganic perovskite compound emerges with improved optical performance and stability [43]. The Goldschmidt tolerance factor t [47] and octahedral factor μ [48] are defined as

$$t = \frac{R_A + R_X}{\sqrt{2}(R_B + R_X)}$$

$$\mu = \frac{R_B}{R_X}$$

R_i , $i = (\text{A}, \text{B}, \text{or X})$ corresponds to the ionic radius of the component element of the perovskite material, ABX_3 . To ensure the stability of perovskite, tolerance factor t , and octahedral factor μ should be in the range $0.813 < t < 1.107$ and $0.442 < \mu < 0.895$ [43].

For PSC design, the absorber layer made of perovskite material is flattened between the ETL and HTL layers. The selection of appropriate ETL and HTL play a significant role in the performance of solar cell. For efficient transportation of charge carriers, affinity (χ) of ETL > should be greater than (χ) of the PSC absorber layer, and ($\chi + E_G$) of HTL < ($\chi + E_G$) of the PSC absorber as demonstrated in Fig. 1.

The operation of light energy conversion into electrical energy is as follows (i) absorption of photon energy, generation of electron-hole pair, and dissociation of excitons (ii) transportation of separated charge carriers towards ETL and HTL (iii) electrodes at the top and bottom of solar cell helps to collect generated charge carriers. In solar cell structure, a layer of perovskite material (intrinsic material) is sandwiched between ETL and HTL, so that light-generated electrons transport toward ETL (n-type material) and holes toward HTL (p-type material).

Solar cell design with machine learning (ML) is an emerging field that has the potential to revolutionize the way of designing and optimization. By combining the power of artificial intelligence (AI) with the principles of physics and materials science, researchers are developing new methods to design solar cells with higher efficiency and lower cost. The performance of solar cells is governed by J_{SC} , V_{OC} , FF and PCE.

ML allows machines to learn from data and enhance performance based on experience, ultimately enabling them to predict output without explicit programming, as defined by Arthur Samuel in 1959. The ML process begins with raw data and involves feature extraction, pre-processing, normalization, training, evaluation, and model deployment [49]. In the field of PV technology, ML can play an important role

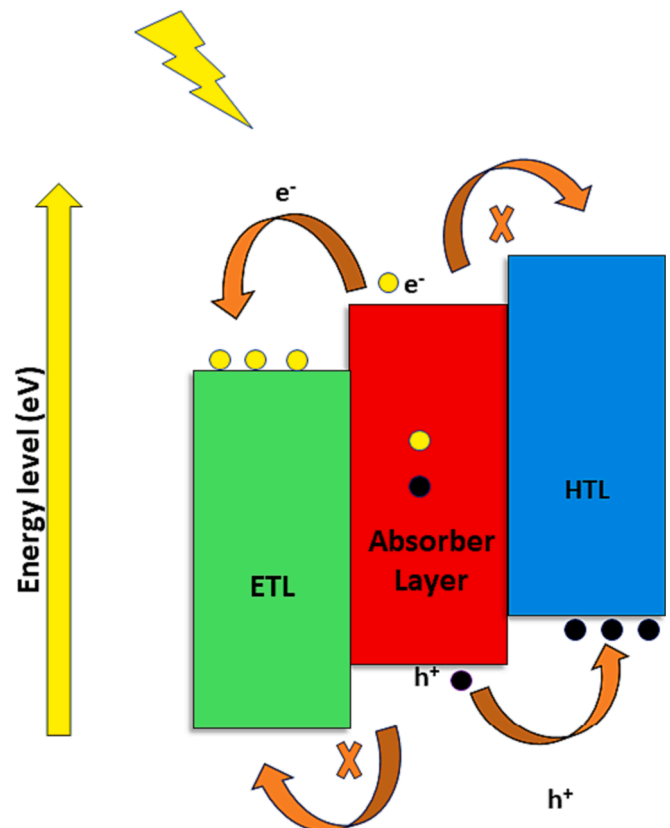


Fig. 1. Transportation of charge carriers toward ETL and HTL.

in the design of solar cells by providing efficient and accurate methods to predict the performance of devices. ML can also be used to optimize the design of solar cell devices by predicting their efficiency and identifying the best optical parameters for their design.

The initial step in ML implementation is to collect quality data. The gathered data can be raw or simulated data from an experiment. By enacting input data attributes with an ML algorithm, a model is designed to predict the output [50]. Fig. 2 depicts the flow of the ML process. ML is categorized as supervised, unsupervised, and reinforcement learning. The most popular deployed ML algorithms are supporting vector machines, decision trees, random forest trees, bayesian networks, regression analysis, genetic algorithms, and artificial neural networks (ANN) [50,51]. Once the model gets trained it takes a short period to predict the performance of solar cells.

This study collaborates the experimental data with AI. With the help of AI, prediction models have been developed and trained from experiment-generated datasets. In the current study, five AI algorithms are implemented to evaluate the performance of ML models. The best-performed ML models, with more accurate predictions, are selected for further analysis. The motive behind this study is to develop ML models, that can forecast the performance of AIHP-SC. Moreover, the training and test data used for ML is collected from the SCAPS-1D (solar cell capacitance simulator-one dimension) simulator. The first phase of the study is dedicated to the simulation of AIHP-SC in the SCAPS-1D simulator. The input data set provided to train the ML model is extracted from the SCAPS-1D simulator, after the simulation of all inorganic $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ based solar cells [43]. By incorporating ML, AI-based models have been designed to predict the PV performance of solar cells that is very much related to the actual performance determined by the simulator.

In this research work, ML models are designed using linear regression, random forest (RF), support vector regression, eXtreme gradient boosting (xGB), and neural networks to predict the PV performance of AIHP-SC. These input attributes applied to ML models, are categorised as training data and testing data. Test data is exclusive from training data, it is used to verify the performance of the model in reference to experimental simulated data.

This work is divided into two parts, firstly AIHP-SC $\text{TiO}_2/\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}/\text{Spiro-MeOTAD}$ is simulated utilizing SCAPS-1D simulator. The PV performance of AIHP-SC is evaluated under the influence of variation in ETL & HTL properties. As abovementioned, the selection of appropriate ETL and HTL has a significant impact on the performance of solar cells. The electron affinity is the key factor, that decides the compatibility of ETL and HTL with absorber layer [52]. Added to this, ETL/absorber layer and absorber layer/HTL interface properties are crucial parameters, that predict the behavior of cells. ETL/HTL mobility and doping density are varied and PV performance of AIHP-SC is obtained at different combinations.

The second phase of the study is dedicated to the implementation of ML, incorporating a data set of 2500 unique combinations, extracted from the simulator. ML models using linear regression, RF, support

vector regression, xGB, and neural network algorithms, to forecast the PV behavior of solar cells while experiencing input data (training data and test data). After designing models, data validation is performed on selected models to verify the performance.

This manuscript is divided into four subsections, continue to this ongoing section 1 of the introduction, device structure and methodology adopted for this investigation are elaborated in section 2. The explanation of extracted results is discussed in section 3 and finally, this study is concluded in section 4 of conclusion.

2. Device structure & methodology

In the current research, the SCAPS-1D simulator is utilized to simulate, all inorganic halide perovskite solar cell (AIHP-SC) layered as $\text{TiO}_2/\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}/\text{Spiro-MeOTAD}$ under one sun illumination. SCAPS-1D simulator is developed by the Department of Electronics and Information Systems (ELIS), University of Gent in Belgium, for PV applications [53]. This simulator accounts for the Poisson equation, different transport models, and continuity equation to evaluate the PV performance parameters of the PV devices [54].

The implementation of a layered solar cell n-i-p structure has ETL of n-type material, which is stacked below the cathode layer (from the front side of the solar cell), and HTL of p-type material placed above the anode (from the rear side). All inorganic perovskite absorber layer $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$, with low lead content, is embedded between ETL and HTL as shown in Fig. 3. The suitable ETL and HTL identified for this solar cell structure are titanium oxide (TiO_2), and Spiro-MeOTAD respectively [55,56]. For this design, the thickness of TiO_2 , $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$, and Spiro-MeOTAD is set to 100 nm, 500 nm, and 150 nm respectively. The thickness of ETL is kept minimum to absorb a maximum of sunlight in $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ [56]. Whereas the absorber layer has been kept optimum thick to maximize/minimize the absorption/recombination. The absorption coefficient utilized for the $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ layer is accessed from SCAPS models and the same

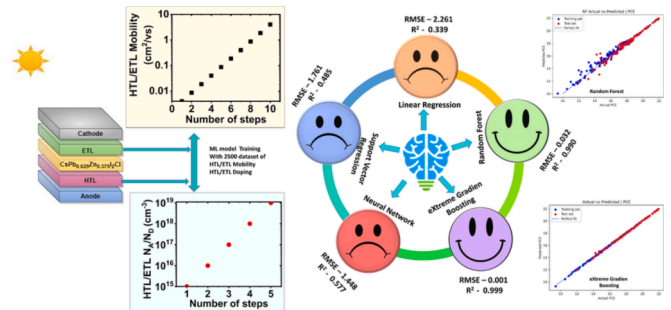


Fig. 3. Device structure and schematic of ML process adopted for $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ based AIHP-SC. All the python codes in jupyter notebook are provided [Supplementary file](#).

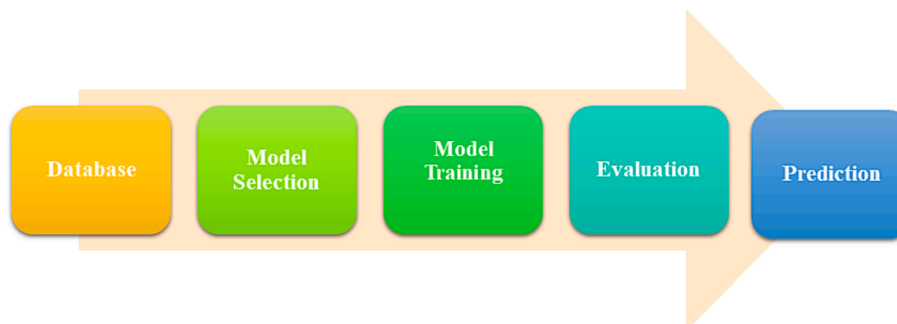


Fig. 2. Process flow of machine learning.

has been represented in Fig. 4. It has been studied that by performing interface engineering, the optical performance of solar cells can be enhanced [57]. Transportation of light-generated charge carriers (in an active layer) toward the electrode mainly relies on properties of ETL, HTL, and their interface (ETL/absorber layer, absorber layer/HTL).

In the foremost part of the research PV performance of solar cell $\text{TiO}_2/\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}/\text{Spiro-MeOTAD}$ is validated under the influence of ETL/HTL mobility, ETL/HTL doping density with the use of SCAPS-1D. ETL mobility is a measure of how quickly generated electrons in the absorber layer are transported toward the electrode. Similarly, HTL mobility determines the mobility of generated holes toward electrodes. Higher charge carrier mobility is required to improve efficiency as well as to avoid recombination. Execution of AIHP-SC at different ETL/HTL mobility (from $0.004 \text{ cm}^2/\text{Vs}$ to $4 \text{ cm}^2/\text{Vs}$) and donor/acceptor doping density ($1 \times 10^{15} \text{ cm}^{-3}$ to $1 \times 10^{19} \text{ cm}^{-3}$) in SCAPS-1D, has generated the data set of 2500 of PV parameters (J_{SC} , V_{OC} , FF and PCE). Subsequently, ML algorithms linear regression, RF, support vector regression, xGB and neural network are implemented on AIHP-SC incorporating simulator generated dataset. The performance of models is analyzed in terms of R^2 and RMSE.

SHAP analysis is also performed in the subsequent part to understand the importance of each parameter (ETL/HTL mobility and donor/acceptor doping density) on the PV performance (J_{SC} , V_{OC} , FF and PCE) of AIHP-SC.

3. Results & discussion

3.1. Simulation using SCAPS-1D simulator

In the first part of this research, the SCAPS-1D simulator has been utilized to realize the impact of ETL & HTL properties on the performance of AIHP-SC. Simulation of $\text{TiO}_2/\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}/\text{Spiro-MeOTAD}$ has generated 2500 result attributes corresponding to different input combinations. This dataset is utilized for feature extraction of the ML model. The material input parameters used for the simulation of solar cell $\text{TiO}_2/\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}/\text{Spiro-MeOTAD}$ in the SCAPS-1D simulator, are mentioned in Table 1. In this simulation, ETL/HTL properties viz. mobility and doping density, are varied over a range to evaluate the PV performance of solar cells. ETL/HTL mobility is varied from $0.004 \text{ cm}^2/\text{Vs}$ to $4 \text{ cm}^2/\text{Vs}$, while the range of ETL/HTL doping density is 10^{15} cm^{-3} to 10^{19} cm^{-3} and a total of 2500 distinct input attributes are generated. Fig. 5 (a) depicts that, after simulation, an insignificant change in J_{SC} has been observed for the entire range of variation corresponding to ETL/HTL mobility, and ETL/HTL doping density. A similar impact has been observed on V_{OC} , and a minor change

in V_{OC} has been noted, after ETL/HTL mobility, ETL/HTL doping density has been varied for the above-mentioned range. The maximum developed V_{OC} is 1.09 V at 10^{19} cm^{-3} of HTL doping density encompassing the entire mentioned range of ETL/HTL mobility, and ETL doping density, depicted in Fig. 5 (b).

On the other hand, Fig. 5(c) illustrates a notable increase in FF from 40.60 % to 88.50 % on variation in ETL/HTL mobility and doping density. At lowest value of ETL/HTL doping density ($1 \times 10^{15} \text{ cm}^{-3}$) and ETL/HTL mobility ($0.004 \text{ cm}^2/\text{Vs}$), extracted FF is 40.7 %. As ETL/HTL doping density and mobility increase, a significant rise in FF is observed. It has been observed that the impact of HTL doping is more prominent on the FF of AIHP-SC in comparison to HTL mobility. It can well be observed from the results that even at lower HTL mobility, an increase in HTL doping leads to a rise in FF from 70.6 % to 88.6 %. However, at low ETL/HTL mobility ($<0.1 \text{ cm}^2/\text{Vs}$) and high doping density ($>1 \times 10^{17} \text{ cm}^{-3}$), the FF of the cell is high ($>82.6 \%$). It has been evaluated that at $0.004 \text{ cm}^2/\text{Vs}$ ETL/HTL mobility and $1 \times 10^{19} \text{ cm}^{-3}$ of ETL/HTL doping density, the resultant FF is 88.50 %. For the same ETL/HTL mobility and keeping ETL doping fixed at $1 \times 10^{19} \text{ cm}^{-3}$, a reduction in HTL doping reduces the FF. However, variation in ETL doping impacts more for range, $1 \times 10^{15} \text{ cm}^{-3}$ to $1 \times 10^{17} \text{ cm}^{-3}$, and at lower ETL mobility (from $0.004 \text{ cm}^2/\text{Vs}$ to $0.186 \text{ cm}^2/\text{Vs}$). Thus, ETL/HTL mobility is less contributing in achieving the highest FF, it is extracted only at maximum ETL/HTL doping ($1 \times 10^{19} \text{ cm}^{-3}$) though ETL/HTL mobility is varied from $0.004 \text{ cm}^2/\text{Vs}$ to $4 \text{ cm}^2/\text{Vs}$.

A noticeable rise in PCE is observed from 9.39 % to 21.90 % while increasing ETL/HTL mobility ($0.004 \text{ cm}^2/\text{Vs}$ to $4 \text{ cm}^2/\text{Vs}$) and ETL/HTL doping density (10^{15} cm^{-3} to 10^{19} cm^{-3}). It is observed from Fig. 5(d), that the highest ETL/HTL doping density (10^{19} cm^{-3}) results maximum PCE of 21.9 %, even though ETL/HTL mobility is at the lower limit ($0.004 \text{ cm}^2/\text{Vs}$). On the other hand, a minimum PCE of 9.39 % is withdrawn, at lower ETL/HTL doping density (10^{15} cm^{-3}) and corresponding ETL/HTL mobility ($0.004 \text{ cm}^2/\text{Vs}$). In addition, the J-V curve and EQE curve correspond to the highest/lowest obtained PCE are plotted in Fig. 6 (a-b). The J_{SC} for both devices is the same and a major reduction in FF is observed therefore an overlapping EQE is obtained for both devices as shown in Fig. 6(b).

In the current study, simulation of $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ based solar cell has concluded that moderation in doping density changed the PCE significantly, however, ETL/HTL mobility has not played any major role.

3.2. Implementation of ML model

In this part of the research, five different ML models are trained with the SCAPS-1D generated data set to predict the optical performance of AIHP-SC. Different ML algorithms are applied to test the accuracy of models. The ML algorithms used in this experiment are linear regression, RF, support vector regression, xGB, and neural networks [66]. linear regression model is a supervised ML model to predicts the linear relation between dependent (predicted value) and independent (actual value) variables [67]. The relationship between dependent and independent variables is described by best fit straight line, which is represented by equation 1.

$$y = x\beta + z \quad (1)$$

where;

y is the dependent variable
x is independent variable
 β is the regression coefficient
z is slope intercept

RF is also the type of supervised learning, that can be implemented as a classifier and regression model. In this, a set of trees is created using a random subset of training data. This is a more impactful learning model, in which the final prediction is concluded based on the average pre-

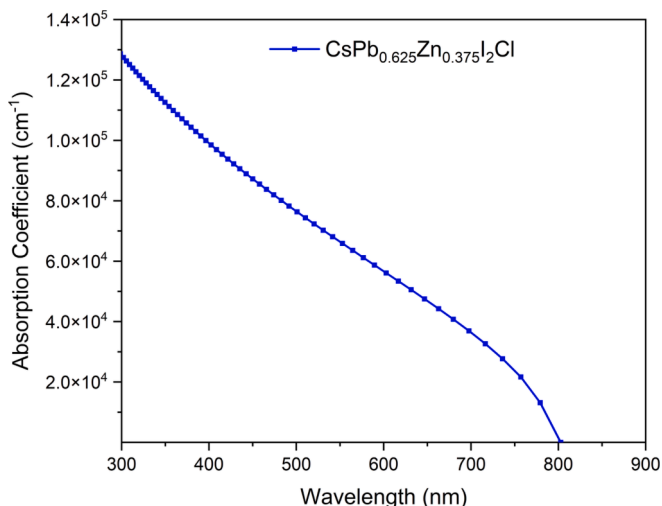
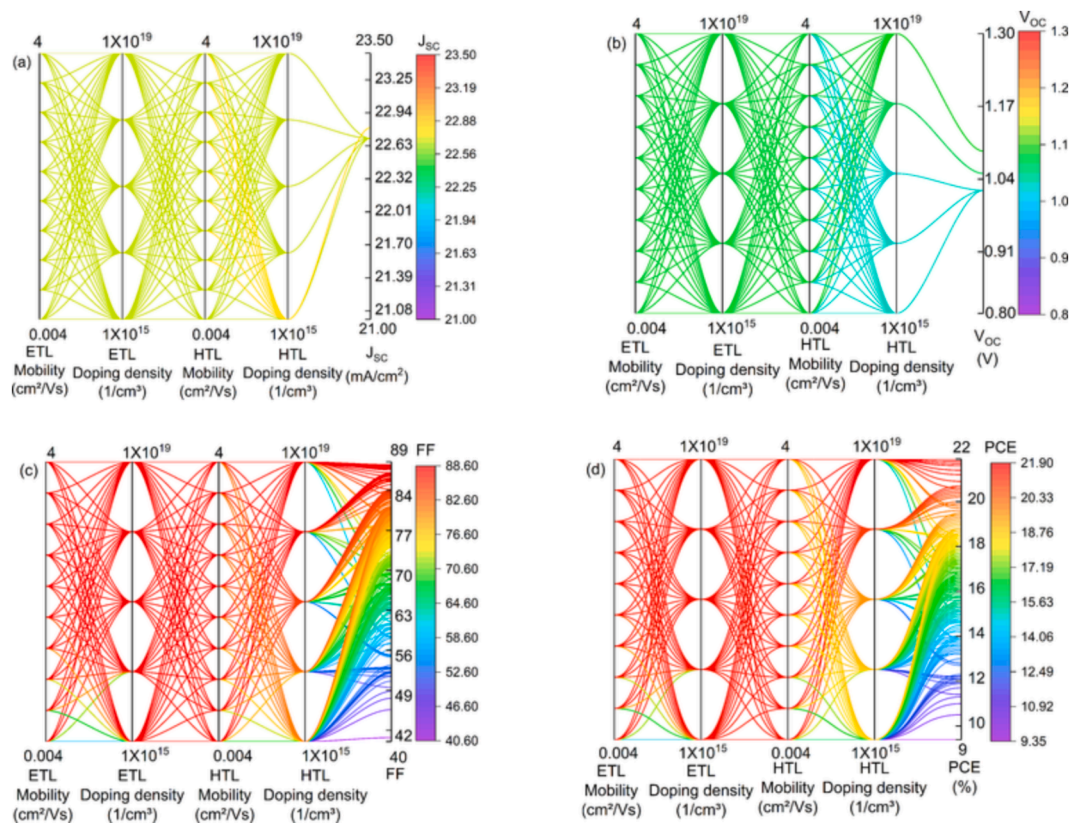


Fig. 4. Absorption coefficient of $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ perovskite layer.

Table 1Material properties of HTL, ETL and CsPb_{0.625}Zn_{0.375}I₂Cl input to SCAPS-1D simulator.

Parameters	Spiro-MeOTAD [58–60]			CsPb _{0.625} Zn _{0.375} I ₂ Cl [43,61–64]	TiO ₂ [59,61,65]
D (nm), thickness	150			500	100
E _g (eV), bandgap	3.0			1.56	3.2
χ (eV), electron affinity	2.45			4.23	4–4.23 (Variable)
ε _r , dielectric permittivity	3.0			6	9
N _C (cm ⁻³), conduction band density	2.2 × 10 ¹⁸			1 × 10 ¹⁹	2.2 × 10 ¹⁸
N _V (cm ⁻³), valence band density	1.8 × 10 ¹⁹			1 × 10 ¹⁹	1.8 × 10 ¹⁹
μ _n (cm ² /Vs), electron mobility	2 × 10 ⁻⁴			25	4
μ _p (cm ² /Vs), Hole mobility	2 × 10 ⁻⁴			25	4
N _a (cm ⁻³), Acceptor density	3 × 10 ¹⁸			1 × 10 ¹⁵	–
N _d (cm ⁻³), Donor density	–			1 × 10 ¹⁵	1 × 10 ¹⁸
Capture cross-section electrons (cm ²)	Defect 1	Defect 2	Defect 3	1 × 10 ⁻¹⁵	1 × 10 ⁻¹²
	1 × 10 ⁻¹⁵	1 × 10 ⁻¹⁷	1 × 10 ⁻¹⁹		
Capture cross-section holes (cm ²)	1 × 10 ⁻¹⁷	1 × 10 ⁻¹⁵	1 × 10 ⁻¹⁸	1 × 10 ⁻¹⁵	1 × 10 ⁻¹²
Energetic distribution	VB tail	CB tail	Gaussian	Single	Single
Energy level w.r.t. reference (eV)	0.6			0.6	1.8
Characteristic energy(eV)	0.01	0.01	0.1	–	–
N _i Total (1/cm ³)	1 × 10 ¹⁴	1 × 10 ¹⁴	1 × 10 ¹⁶	1 × 10 ¹⁴	1 × 10 ¹⁹
N _i peak (1/eV/cm ³)	1 × 10 ¹⁶	1 × 10 ¹⁶	5.642 × 10 ¹⁶	–	–

For all the materials- The defect type is neutral, reference for defect energy level E_t is above E_v**Fig. 5.** Impact of ETL/HTL mobility and doping density variation on (a) J_{sc} (b) V_{oc} (c) FF (d) PCE. Complete datasets are provided in [Supplementary file](#).

diction of each tree [68].

$$f(x) = a_0(x) + a_1(x) + a_2(x) + \dots + a_n(x) \quad (2)$$

Another ML model is support vector regression, it is a type of regression, in which a function is required to be evaluated, which estimates the relation between input variables and target variables and minimizes the prediction error (difference between input variable and target variable). This regression model aims to map the input variables to the desired target variables, by using training data [69].

Another widely used ML model is eXtreme gradient boosting (xGB),

which is a decision tree ML model, that deploys a gradient descent function, to predict the output. This model works both on gradient boosting and decision trees. This algorithm is scalable as well as highly accurate. xGB performs efficiently, with high speed on large datasets [70,71].

Further, ANN is a model of deep learning, which acts as a human brain to generate the prediction. ANN is a model of interconnected nodes, similar to a human brain set of interconnected neurons. Data is being processed by three types of layers - an input layer, hidden layers, and output layers. The data set and feature set are applied at input nodes [72]. The data entered into hidden nodes are multiplied with predefined

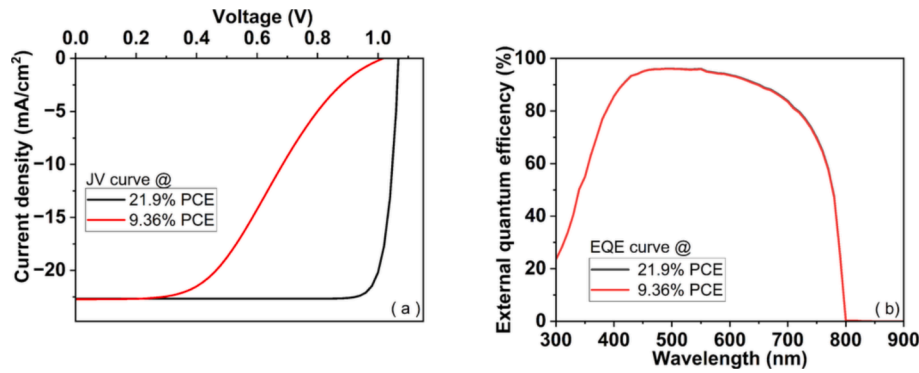


Fig. 6. The resultant plots of (a) J-V and (b) EQE correspond to the highest/lowest extracted PCE. There are 299 combinations where highest efficiency of 21.9% is achieved (please refer to [Supplementary file](#)).

weights, summation of these outputs generates a single output.

The data set input to the ML model is a combination of features (independent variables) and output of AIHP-SC (dependent variable). The material parameters are considered features of the ML model. This data set is further split into an 80:20 ratio, 80 % of the data is being used to train the ML model, and the remaining 20 % is considered as test data. Once the ML model gets trained with a data set (combination of features and output parameters of PSCs), test data is applied to the ML model to verify the performance of AIHP-SC. In this research, we have applied linear regression, RF, support vector regression, xGB, and neural network ML models on a dataset. Error metrics average root mean square error (RMSE) and R-squared (R^2) determined the accuracy of these ML models.

The data set is generated through the SCAPS-1D simulator, as discussed earlier. The AIHP-SC attributes ETL/HTL mobility and doping density as well SCAPS- 1D generated PV performance are utilized to generate prediction of PV performance. This ML-based investigation, where V_{OC} , J_{SC} , FF, and PCE are considered as dependent target variables and ETL/HTL mobility and doping density as independent variables. The ML model using RF and the xGB algorithm, have generated highly accurate prediction which is very close to simulated results. The value of average RMSE and R^2 evaluate the performance of the regression model, it indicates how closely the prediction generated by the ML model is matched to the actual performance of AIHP-SC. The Obtained average RMSE and R^2 for five ML algorithms are mentioned in [Table 2](#).

Ideal average root mean square error (RMSE) – 0 and R-squared (R^2) – 1.

It has been analyzed that higher prediction accuracy is realized at smaller values of RMSE i.e., closer to 0. In this study, average RMSE/ R^2 values obtained from RF and xGB algorithms are 0.032/0.990 and 0.001/0.999 respectively. It can be observed from the resultant RMSE and R^2 values of the xGB and RF model, that predictions generated by these two algorithms are highly accurate.

In relation to verifying the mapping of training data with prediction

response, scatter plots are utilized. The response of both RF-based and xGB-based ML models has been represented by [Fig. 7](#), where notations (i) and (ii) are used for RF and xGB models respectively. The response of RF and xGB-based ML model is analyzed by evaluated output PV parameters (J_{SC} , V_{OC} , FF and PCE) of PSC. For this work, the target variables under consideration are J_{SC} , V_{OC} , FF and PCE (PV parameters), and ML models are trained to map the input parameters into target variables. It has been well observed from the extracted results that actual (simulator-generated) and predicted outputs (ML model-generated) are closely matched using RF and xGB algorithms. With the help of a scatter plot correlation between training and testing has been analyzed. The blue and red dots shown in scatter plots represent training data and testing data respectively. [Fig. 7a\(i\)](#) and [Fig. 7a\(ii\)](#) illustrate the relationship between input training attributes and predicted values of J_{SC} , generated by the RF and xGB algorithms respectively. It has been concluded from the plot that the test dataset of J_{SC} (dependent variable) is appropriately mapped to the target variable. The scatter plot of J_{SC} obtained from the RF-based ML model and xGB-based ML model are the best-fit models, which depict that the prediction generated by both models is very much nearer to actual values.

Similarly, predicted V_{OC} extracted from RF-based ML model/xGB signifies the high accuracy in a correlation between actual and predicted value, depicted in [Fig. 7b\(i\)/7b\(ii\)](#). From [Fig. 7c\(i\)/7c\(ii\)](#) and [Fig. 7d\(i\)/7d\(ii\)](#), it has been analyzed that the response generated by the xGB ML model, in terms of FF and PCE shows a high accuracy compared to prediction extracted from RF-based ML model. It can be visualized from [Fig. 7c\(ii\)](#) that the scatter plot of FF is mapped perfectly on the fit line with the help of an xGB-based ML model. Whereas, RF-based ML model-generated response indicates the more accurate matching in predicted and actual FF at the upper range relative to a lower range of FF indicated in [Fig. 7c\(ii\)](#). The predicted PCE obtained from the xGB model is perfectly matched to the simulator-generated PCE, illustrated in [Fig. 7d\(ii\)](#). It has been well observed from average RMSE, R^2 and scatter plots, that prediction generated by xGB based ML model is better than the RF-based ML model. Therefore, based on the investigation findings, it has been concluded that RF and xGB-based ML models are the optimal choice to predict the response of AIHP-SC.

To insight high accuracy in the prediction model, the K fold cross-validation process is incorporated. The entire data set includes train and test data, which are split into non-overlapped K folds of equal size. In cross-validation, the process is repeated K times, and in each iteration, one-fold is reserved as test data remaining folds as train data for the ML model. In this way, each fold acts as test data once and evaluates the performance of the ML model. This process is completed until all folds are considered as test folds. In this research 50-fold cross-validation is set, according to the algorithm process repeated 50 times to predict the accuracy of ML model.

Table 2

Comparative analysis of obtained error metrics (root mean square error and R-squared) of ML algorithms.

Error Metrics	AI Algorithms				
	Linear Regression	Random Forest	Support Vector Regression	eXtreme gradient boosting	Neural Network
Root Mean Squared Error (RMSE)	2.261	0.032	1.761	0.001	1.448
R-squared (R^2)	0.339	0.990	0.485	0.999	0.577

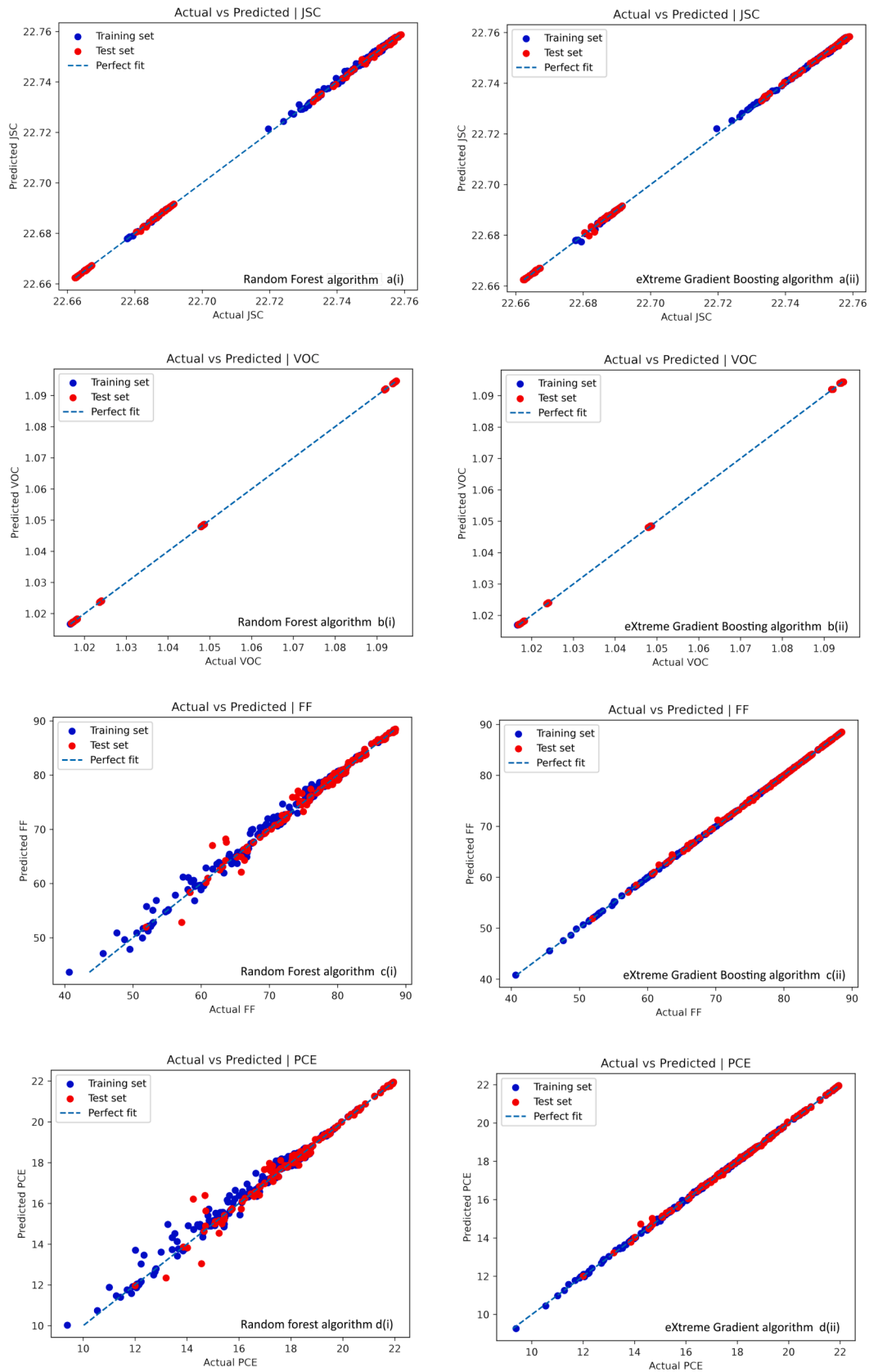


Fig. 7. Correlation between predicted and actual PV performance of AIHP-SC obtained through RF /xGB model; a(i)/(ii) J_{SC} , b(i)/(ii) V_{OC} , c(i)/(ii) FF and d(i)/(ii) PCE.

3.3. Feature Significance

This section of the investigation validates the influence of features (ETL and HTL parameters) on the performance of the adopted ML model. SHAP analysis is performed to identify the impact of features on the output of the trained ML model. It is an indicative measure to identify how much a particular feature contributes to the final prediction of the ML model. As discussed, in the previous section, target variables (output performance parameters of ML model) considered for this investigation are J_{SC} , V_{OC} , FF and PCE (also known as dependent variables) and features which has predicted the behavior of AIHP-SC are ETL/ HTL mobility and doping.

After SHAP was implemented on the RF-based ML model and xGB-based models, it has been well analyzed that the impact of HTL doping on J_{SC} is more prominent, as shown in Fig. 8a(i) and 8a(ii) respectively. Similarly, V_{OC} is strongly influenced by HTL doping, as represented in Fig. 8b(i) and 8b(ii).

Whereas, both the RF-based ML model and xGB-based models predicted that ETL mobility, ETL doping and HTL doping influenced more on FF and PCE of AIHP-SC, as portrayed in Fig. 8(c) Fig. and 8(d). Therefore, it is recommended to optimize these three features- ETL mobility, and ETL/HTL doping density to improve the PCE of AIHP-SC.

4. Conclusion

This research work proposed an approach to integrating artificial intelligence in photovoltaic devices. With the use of SCAPS-1D simulator AIHP-SC has been designed under one sun illumination AM1.5G

spectrum. A low lead all inorganic perovskite active layer $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ utilized in solar cell design improved the thermal stability as well the optical performance. Various ML models viz., linear regression, RF, support vector regression, xGB, and neural network algorithms have been employed in this research work, to generate a prediction of AIHP-SC. The supportive training and test data for ML algorithms have been generated through the SCAPS-1D simulator. The performance of the abovementioned ML models has been evaluated with the help of a dataset of 2500 attributes. Based on evaluated MSE and R^2 , RF and xGB outweigh the remaining implemented models. The MSE obtained from RF and xGB algorithms is 0.032, and 0.001 respectively. And estimated R^2 is 0.990/0.999 for RF/xGB algorithms. Evaluated R^2 and MSE metrics, permitting consideration of RF and xGB models to generate a prediction. Performance prediction generated by RF and xGB models at different combinations of ETL/HTL mobility and doping density results in high accuracy with the SCAPS-1D data set (actual data). In the subsequent phase of the study, the influence of the independent variable on the target variable is verified with the implementation of SHAP analysis. It has been found in the results that J_{SC} and V_{OC} are strongly influenced by HTL doping. Whereas, the independent variable ETL/HTL doping and ETL mobility contributed more to the prediction of FF and PCE. This work concluded that RF and xGB algorithms perform best to generate a prediction for $\text{CsPb}_{0.625}\text{Zn}_{0.375}\text{I}_2\text{Cl}$ based AIHP-SC. The incorporation of artificial intelligence in AIHP-SC design leads pathway for researchers to generate performance prediction with minimum resources before actual design.

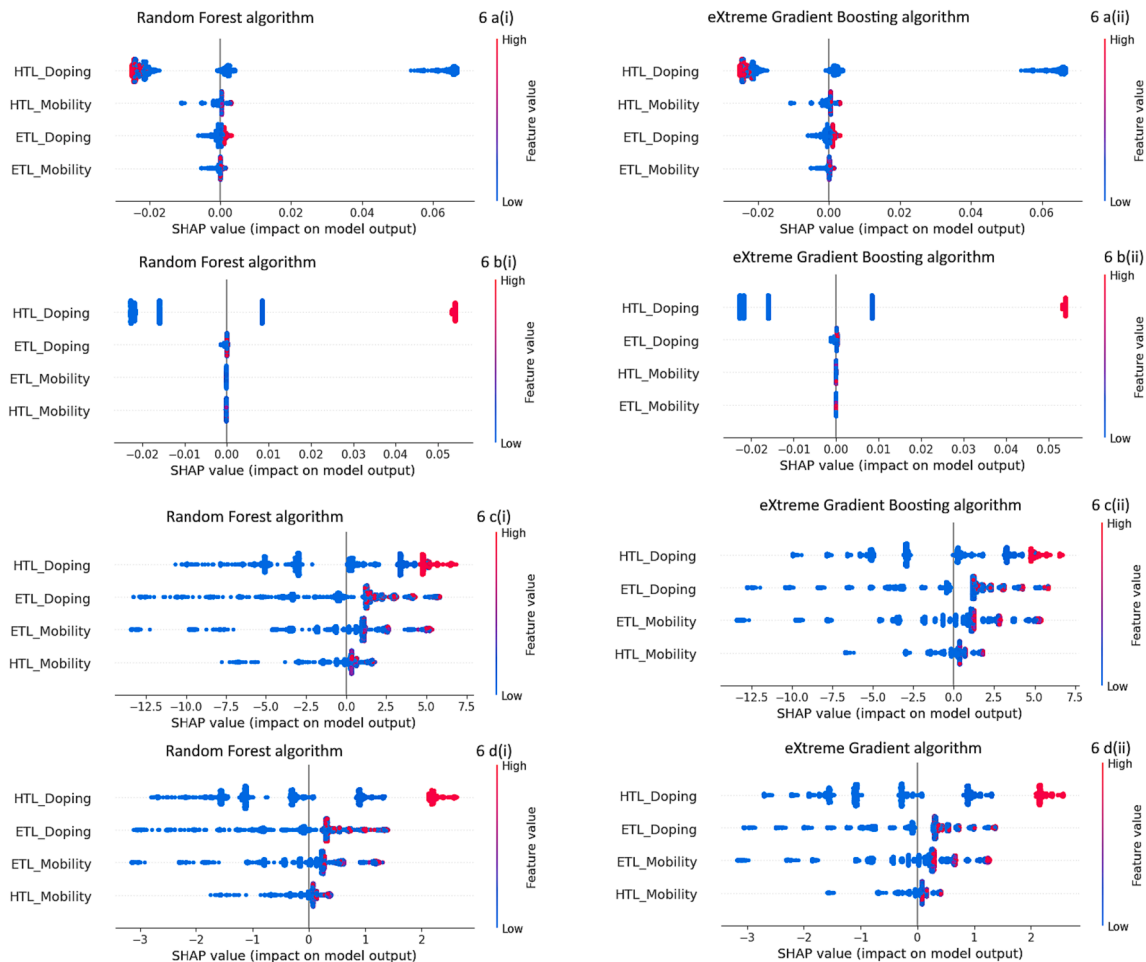


Fig. 8. SHAP analysis performed for RF/xGB model; a(i)/(ii) J_{SC} , b(i)/(ii) V_{OC} , c(i)/(ii) FF and d(i)/(ii) PCE.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful to Dr. Marc Burgelman and his staff, at the University of Gent, Belgium, for providing the open-source SCAPS-1D simulator.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.solener.2023.112055>.

References

- Tai, Q., Yan, F., 2017. Emerging semitransparent solar cells: materials and device design. *Adv. Mater.* 29 (34), 1700192.
- Bhattarai, S., Pandey, R., Madan, J., Mhamdi, A., Bouazizi, A., Muchahary, D., Gogoi, D., Sharma, A., Das, T.D., 2022. Investigation of carrier transport materials for performance assessment of lead-free perovskite solar cells. *IEEE Trans. Electron Devices* 69 (6), 3217–3224.
- Mutalikdesai, A., Ramasesha, S.K., 2017. Emerging solar technologies: Perovskite solar cell. *Resonance* 22 (11), 1061–1083.
- Maes, J., Balcaen, L., Drijvers, E., Zhao, Q., De Roo, J., Vantomme, A., Vanhaecke, F., Geiregat, P., Hens, Z., 2018. Light absorption coefficient of CsPbBr₃ perovskite nanocrystals. *J. Phys. Chem. Lett.* 9 (11), 3093–3097.
- Ifitkhar, F.J., et al., 2021. Structural and optoelectronic properties of hybrid halide perovskites for solar cells. *Org. Electron.* 91, 106077.
- Pistor, P., Ruiz, A., Cabot, A., Izquierdo-Roca, V., 2016. Advanced Raman spectroscopy of methylammonium lead iodide: development of a non-destructive characterisation methodology. *Sci. Rep.* 6 (1).
- Shrivastav, N., Madan, J., Pandey, R., Shalan, A.E., 2021. Investigations aimed at producing 33% efficient perovskite-silicon tandem solar cells through device simulations. *RSC Adv.* 11 (59), 37366–37374.
- Wang, B., Xiao, X., Chen, T., 2014. Perovskite photovoltaics: a high-efficiency newcomer to the solar cell family. *Nanoscale* 6 (21), 12287–12297.
- Kojima, A., Teshima, K., Miyasaka, T., Shirai, Y., 2006. Novel Photoelectrochemical Cell with Mesoscopic Electrodes Sensitized by Lead-Halide Compounds (2). *ECS Meeting Abstracts* MA2006-02 (7), 397.
- Kojima, A., Teshima, K., Shirai, Y., Miyasaka, T., 2009. Organometal halide perovskites as visible-light sensitizers for photovoltaic cells. *J. Am. Chem. Soc.* 131 (17), 6050–6051.
- Jeong, J., Kim, M., Seo, J., Lu, H., Ahlawat, P., Mishra, A., Yang, Y., Hope, M.A., Eickemeyer, F.T., Kim, M., Yoon, Y.J., Choi, I.W., Darwich, B.P., Choi, S.J., Jo, Y., Lee, J.H., Walker, B., Zakeeruddin, S.M., Emsley, L., Rothlisberger, U., Hagfeldt, A., Kim, D.S., Grätzel, M., Kim, J.Y., 2021. Pseudo-halide anion engineering for α -FAPbI₃ perovskite solar cells. *Nature* 592 (7854), 381–385.
- Jayan, K.D., Sebastian, V., 2021. Comprehensive device modelling and performance analysis of MASnI₃ based perovskite solar cells with diverse ETM, HTM and back metal contacts. *Sol. Energy* 217, 40–48.
- Pandey, R., Bhattarai, S., Sharma, K., Madan, J., Al-Mousoi, A.K., Mohammed, M. K.A., Hossain, M.K., 2023. Halide Composition Engineered a Non-Toxic Perovskite-Silicon Tandem Solar Cell with 30.7% Conversion Efficiency. *ACS Appl. Electron. Mater.* <https://doi.org/10.1021/acsaelm.2c01574>.
- Tara, A., et al., 2021. Device simulation of FASnI₃ based perovskite solar cell with Zn (00. 3, S0. 7) as electron transport layer using SCAPS-1D. *Opt. Mater.* 119, 111362.
- Nie, W., Blancon, J.-C., Neukirch, A.J., Appavoo, K., Tsai, H., Chhowalla, M., Alam, M.A., Feir, M.Y., Katan, C., Even, J., Tretiak, S., Crochet, J.J., Gupta, G., Mohite, A.D., 2016. Light-activated photocurrent degradation and self-healing in perovskite solar cells. *Nat. Commun.* 7 (1).
- Chen, H., Zhou, F., Jin, Z., 2021. Interface engineering, the trump-card for CsPbX₃ (X = I, Br) perovskite solar cells development. *Nano Energy* 79, 105490.
- Pathania, A., et al., 2020. Design and optimization of 26.3% efficient perovskite/FeSi 2 monolithic tandem solar cell. *J. Mater. Sci. Mater. Electron.* 31, 15218–15224.
- Pandey, R., Chaujar, R., 2017. Technology computer aided design of 29.5% efficient perovskite/interdigitated back contact silicon heterojunction mechanically stacked tandem solar cell for energy-efficient applications. *J. Photonics for Energy*, 7 (2), 022503. <https://doi.org/10.1117/1.JPE.7.022503>.
- Bella, F., Renzi, P., Cavallo, C., Gerbaldi, C., 2018. *Caesium for perovskite solar cells: an overview*. Chemistry—A European Journal 24 (47), 12183–12205.
- Chen, Y., Jing, H., Ling, F., Kang, W., Zhou, T., Liu, X., Zeng, W., Zhang, Y., Qi, L., Fang, L., Zhou, M., 2019. Tuning the electronic structures of all-inorganic lead halide perovskite CsPbI₃ via heterovalent doping: A first-principles investigation. *Chem. Phys. Lett.* 722, 90–95.
- Huang, J., Yuan, Y., Shao, Y., Yan, Y., 2017. Understanding the physical properties of hybrid perovskites for photovoltaic applications. *Nat. Rev. Mater.* 2 (7).
- Bahiraie, M., Godini, A., Shahsavari, A., 2018. Thermal and hydraulic characteristics of a minichannel heat exchanger operated with a non-Newtonian hybrid nanofluid. *J. Taiwan Inst. Chem. Eng.* 84, 149–161.
- Liang, J., Wang, C., Wang, Y., Xu, Z., Lu, Z., Ma, Y., Zhu, H., Hu, Y.I., Xiao, C., Yi, X. u., Zhu, G., Lv, H., Ma, L., Chen, T., Tie, Z., Jin, Z., Liu, J., 2016. All-inorganic perovskite solar cells. *J. Am. Chem. Soc.* 138 (49), 15829–15832.
- Yao, Z., Zhao, W., Liu, S.F., 2021. Stability of the CsPbI₃ perovskite: From fundamentals to improvements. *J. Mater. Chem. A* 9 (18), 11124–11144.
- Liang, J., Zhao, P., Wang, C., Wang, Y., Hu, Y.I., Zhu, G., Ma, L., Liu, J., Jin, Z., 2017. CsPb0.9Sn0.1Br₂ based all-inorganic perovskite solar cells with exceptional efficiency and stability. *J. Am. Chem. Soc.* 139 (40), 14009–14012.
- Chang, X., Li, W., Zhu, L., Liu, H., Geng, H., Xiang, S., Liu, J., Chen, H., 2016. Carbon-based CsPbBr₃ perovskite solar cells: all-ambient processes and high thermal stability. *ACS Appl. Mater. Interfaces* 8 (49), 33649–33655.
- Yang, H., et al., Recent Progress of Film Fabrication Process for Carbon-Based All-Inorganic Perovskite Solar Cells. *Crystals*, 2023. 13(4): p. 679.
- Hossain, M.K., Mohammed, M.K.A., Pandey, R., Arnab, A.A., Rubel, M.H.K., Hossain, K.M., Ali, M.H., Rahman, M.F., Bencherif, H., Madan, J., Islam, M.R., Samajdar, D.P., Bhattarai, S., 2023. Numerical Analysis in DFT and SCAPS-1D on the Influence of Different Charge Transport Layers of CsPbBr₃ Perovskite Solar Cells. *Energy Fuel* 37 (8), 6078–6098.
- Wang, J., et al., 2023. 21.15%-Efficiency and Stable γ -CsPbI₃ Perovskite Solar Cells Enabled by an Acyloin Ligand. *Adv. Mater.* 2210223.
- Idrissi, S., et al., 2021. Band gaps of the lead perovskites photovoltaic CsXCl₃ (X = Sn, Pb or Ge). *Mater. Sci. Semicond. Process.* 122, 105484.
- Yuan, H., Zhao, Y., Duan, J., He, B., Jiao, Z., Tang, Q., 2018. Enhanced charge extraction by setting intermediate energy levels in all-inorganic CsPbBr₃ perovskite solar cells. *Electrochim. Acta* 279, 84–90.
- Li, J., Cao, H.-L., Jiao, W.-B., Wang, Q., Wei, M., Cantone, I., Lü, J., Abate, A., 2020. Biological impact of lead from halide perovskites reveals the risk of introducing a safe threshold. *Nat. Commun.* 11 (1).
- Giustino, F., Snaith, H.J., 2016. Toward lead-free perovskite solar cells. *ACS Energy Lett.* 1 (6), 1233–1240.
- Prakash, J., Singh, A., Sathiyar, G., Ranjan, R., Singh, A., Garg, A., Gupta, R.K., 2018. Progress in tailoring perovskite based solar cells through compositional engineering: Materials properties, photovoltaic performance and critical issues. *Mater. Today Energy* 9, 440–486.
- Eperon, G.E., Paternò, G.M., Sutton, R.J., Zampetti, A., Haghighirad, A.A., Cacaili, F., Snaith, H.J., 2015. Inorganic caesium lead iodide perovskite solar cells. *J. Mater. Chem. A* 3 (39), 19688–19695.
- Lin, P.-Y., Loganathan, A., Raifuku, I., Li, M.-H., Chiu, Y.-Y., Chang, S.-T., Fakharuddin, A., Lin, C.-F., Guo, T.-F., Schmidt-Mende, L., Chen, P., 2021. Pseudo-halide perovskite solar cells. *Adv. Energy Mater.* 11 (28).
- Pathania, A., Madan, J., Pandey, R., Sharma, R., 2020. Effect of structural and temperature variations on perovskite/Mg 2 Si based monolithic tandem solar cell structure. *Appl. Phys. A* 126 (7).
- Calvo, M.E., 2017. Materials chemistry approaches to the control of the optical features of perovskite solar cells. *J. Mater. Chem. A* 5 (39), 20561–20578.
- Wang, G., Lei, M., Liu, J., He, Q., Zhang, W., 2020. Improving the Stability and Optoelectronic Properties of All Inorganic Less-Pb Perovskites by B-Site Doping for High-Performance Inorganic Perovskite Solar Cells. *Solar RRL* 4 (12).
- Shen, X., Zhang, Y., Kershaw, S.V., Li, T., Wang, C., Zhang, X., Wang, W., Li, D., Wang, Y., Lu, M., Zhang, L., Sun, C., Zhao, D., Qin, G., Bai, X., Yu, W.W., Rogach, A.L., 2019. Zn-alloyed CsPbI₃ nanocrystals for highly efficient perovskite light-emitting devices. *Nano Lett.* 19 (3), 1552–1559.
- Sa, R., Wang, L., Wang, J., Chen, X., Chen, J., Lou, B., 2019. First-principles insights of electronic and optical properties of Zn-doped CH₃NH₃PbI₃ for photovoltaic applications. *Appl. Phys. Express* 12 (8), 082011.
- Khan, U., Zhinong, Y., Khan, A.A., Zulfikar, A., Ullah, N., 2019. High-performance CsPbI₂Br perovskite solar cells with zinc and manganese doping. *Nanoscale Res. Lett.* 14 (1).
- Chang, J., Wu, Q.I., Gao, C.-H., Huang, Y., Ju, M., Wang, G., Yuan, H., Chen, H., 2022. A Hybrid Functional Study on Perovskite-Based Compounds CsPb1– α Zn α I3– β X β (X = Cl or Br). *J. Phys. Chem. Lett.* 13 (25), 5900–5909.
- Noh, J.H., Im, S.H., Heo, J.H., Mandal, T.N., Seok, S.I., 2013. Chemical management for colorful, efficient, and stable inorganic-organic hybrid nanostructured solar cells. *Nano Lett.* 13 (4), 1764–1769.
- Kumawat, N.K., Dey, A., Gopinathan, S.P., Narasimhan, K.L., Kabra, D., 2015. Band gap tuning of CH₃NH₃Pb (Br 1–x Cl x) 3 hybrid perovskite for blue electroluminescence. *ACS Appl. Mater. Interfaces* 7 (24), 13119–13124.
- Zhang, Y., Wu, C., Wang, D., Zhang, Z., Qi, X., Zhu, N., Liu, G., Li, X., Hu, H., Chen, Z., Xiao, L., Qu, B.O., 2019. High efficiency (16.37%) of cesium bromide-passivated all-inorganic CsPbI₂Br perovskite solar cells. *Solar RRL* 3 (11).
- Goldschmidt, V.M., 1926. Die gesetze der krystallochemie. *Naturwissenschaften* 14 (21), 477–485.
- Li, C., Lu, X., Ding, W., Feng, L., Gao, Y., Guo, Z., 2008. Formability of abx₃ (x = f, cl, br, i) halide perovskites. *Acta Crystallogr. B* 64 (6), 702–707.
- Mellit, A. and S. Kalogirou, *Machine learning and deep learning for photovoltaic applications, in Artificial Intelligence for Smart Photovoltaic Technologies*. 2022, AIP Publishing LLC Melville, New York. p. 1-1-1-20.

- [50] Tr, R., Lilhore, U.K., M, P., Simaiya, S., Kaur, A., Hamdi, M., 2022. Predictive analysis of heart diseases with machine learning approaches. *Malaysian Journal of Computer Science* 132–148.
- [51] Bansal, M., Goyal, A., Choudhary, A., 2022. A comparative analysis of K-Nearest Neighbour, Genetic, Support Vector Machine, Decision Tree, and Long Short Term Memory algorithms in machine learning. *Decision Analytics Journal*, 100071.
- [52] Singh, N., Agarwal, A., Agarwal, M., 2022. Study the effect of band offsets on the performance of lead-free double perovskite solar cell. *Opt. Mater.* 125, 112112.
- [53] Burgelman, M., et al., *SCAPS manual*. February, 2016.
- [54] Burgelman, M., Nollet, P., Degrave, S., 2000. Modelling polycrystalline semiconductor solar cells. *Thin Solid Films* 361, 527–532.
- [55] Kavan, L., 2019. Conduction band engineering in semiconducting oxides (TiO₂, SnO₂): Applications in perovskite photovoltaics and beyond. *Catal. Today* 328, 50–56.
- [56] Sharma, S., et al., 2021. Numerical simulation and proof of concept for performance assessment of cesium based lead-free wide-bandgap halide solar cells. *Opt. Mater.* 111, 110644.
- [57] Shadrokh, Z., et al., 2020. Enhanced photovoltaic performance and stability of perovskite solar cells by interface engineering with poly (4-vinylpyridine) and Cu₂ZnSnS₄&CNT. *Sol. Energy* 201, 908–915.
- [58] Pandey, R., Singla, A., Madan, J., Sharma, R., Chaujar, R., 2019. Toward the design of monolithic 23.1% efficient hysteresis and moisture free perovskite/c-Si HJ tandem solar cell: a numerical simulation study. *J. Micromech. Microeng.* 29 (6), 064001.
- [59] Nakanishi, A., Takiguchi, Y., Miyajima, S., 2016. Device simulation of CH₃NH₃PbI₃ perovskite/heterojunction crystalline silicon monolithic tandem solar cells using an n-type a-Si:H/p-type μ c-Si 1-x O x :H tunnel junction : Device simulation of perovskite/heterojunction c-Si monolithic tandem solar cells. *Phys. Status Solidi A* 213 (7), 1997–2002.
- [60] Bhattarai, S., Das, T., 2021. Optimization of carrier transport materials for the performance enhancement of the MAGel3 based perovskite solar cell. *Sol. Energy* 217, 200–207.
- [61] Yue, M., Su, J., Zhao, P., Lin, Z., Zhang, J., Chang, J., Hao, Y., 2019. Optimizing the performance of CsPbI₃-based perovskite solar cells via doping a ZnO electron transport layer coupled with interface engineering. *Nano-Micro Letters* 11 (1).
- [62] Jayan K, D., Sebastian, V., Kurian, J., 2021. Simulation and optimization studies on CsPbI₃ based inorganic perovskite solar cells. *Sol. Energy* 221, 99–108.
- [63] Yang, Z., Surrente, A., Galkowski, K., Miyata, A., Portugall, O., Sutton, R.J., Haghighirad, A.A., Snaith, H.J., Maude, D.K., Plochocka, P., Nicholas, R.J., 2017. Impact of the halide cage on the electronic properties of fully inorganic cesium lead halide perovskites. *ACS Energy Lett.* 2 (7), 1621–1627.
- [64] Singh, R.K., Kumar, R., Jain, N., Dash, S.R., Singh, J., Srivastava, A., 2019. Investigation of optical and dielectric properties of CsPbI₃ inorganic lead iodide perovskite thin film. *J. Taiwan Inst. Chem. Eng.* 96, 538–542.
- [65] Rai, N., et al., 2020. Analysis of various ETL materials for an efficient perovskite solar cell by numerical simulation. *J. Mater. Sci. Mater. Electron.* 31, 16269–16280.
- [66] Yilmaz, B., Yildirim, R., 2021. Critical review of machine learning applications in perovskite solar research. *Nano Energy* 80, 105546.
- [67] Maulud, D., Abdulazeez, A.M., 2020. A review on linear regression comprehensive in machine learning. *J. Appl. Sci. Technol. Trends* 1 (4), 140–147.
- [68] Chen, R.-C., Dewi, C., Huang, S.-W., Caraka, R.E., 2020. Selecting critical features for data classification based on machine learning methods. *Journal of Big Data* 7 (1).
- [69] Gholami, R., Fakhari, N., 2017. Support vector machine: principles, parameters, and applications. In: *Handbook of Neural Computation*. Elsevier, pp. 515–535.
- [70] Chen, T., Guestrin, C., 2016. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*.
- [71] Kukreja, V., Dhiman, P., 2020. A Deep Neural Network Based Disease Detection Scheme for Citrus Fruits. 2020 International conference on smart electronics and communication (ICOSEC). IEEE.
- [72] Zou, J., Han, Y., So, S.-S., 2009. Overview of artificial neural networks. *Artificial Neural Networks: Methods Appl.* 14–22.